

Statement of Work (SOW)

Project Title: DNA extraction and next-generation genotyping services for 384 Maize leaf samples

Description

The USDA ARS Corn Host Plant Resistance Research Unit requires DNA extraction and next-generation sequencing services for 384 maize leaf samples. ARS scientists will provide freeze-dried maize leaf tissue organized in 96-well microtiter plates for processing. The Contractor shall perform all required services, including DNA extraction, library preparation, sequencing, and data analysis. The resulting SNP data will support downstream genetic, breeding, diversity, and genome-wide association studies. This work will contribute to the development of improved maize inbred lines and advance our research objectives under NP301: Plant Genetic Resources, Genomics, and Genetic Improvement.

Performance Requirements

- Extract high-quality genomic DNA from freeze-dried maize leaf tissue of variable quality. DNA concentration and purity must meet the minimum standards required for genotyping platforms. All DNA quality-control (QC) metrics must be documented and provided to the Government. Samples that fail QC must be identified and reported. A minimum of 90 percent of all samples must meet the required DNA quality standards for genotyping.
- Prepare genotyping by sequencing, or next-generation genotyping libraries using approved, commercially available methods employing the ApeKI restriction enzyme. A PCR-free preparation method must be used to construct a specialized library. The Contractor must include both positive and negative controls during library preparation. Negative controls consist of water used during library preparation, and positive controls consist of DNA from known reference samples.
- Next Generation Sequencing must be completed using the Illumina NovaSeq X 10B (300 cycles shared flow cell) or similar platform with a read length of 2 x 150 (Paired End), with a minimum sequencing depth of 25× per sample to support marker discovery. All samples must be run together in a single batch to maintain consistency and ensure optimal data quality
- Ensure the generation of high-quality genotyping data with acceptable call rates. Each successfully genotyped sample must achieve a call rate of at least 95 percent. Clean reads must align to the B73 RefGen_v5 reference genome assembly. A minimum of 15,000 SNP markers with known chromosomal positions, distributed across all maize chromosomes, must be produced.
- Deliver user-friendly genotype data files with descriptive column headers, including sample IDs, marker names, chromosomal positions, and genotype calls. Provide both raw

sequencing data and processed genotype data in VCF format on a hard disk. All data generated must be handled confidentially and released only to authorized project personnel.

- Complete all processing activities within 15 weeks of sample receipt.
- Provide regular status updates to the Government throughout the performance period.